A G O R A

A sandbox for the economics of abundance

Who Owns the Machine?

AI, the labour share and Europe's policy choices

What happens to Europe's households and public finances if AI moves income from work to capital, and what cash transfers and citizen ownership can and cannot do about it. Tested in a stock-flow-consistent model of 24 EU economies.

A sandbox, not an oracle. Please read this first. Every number here is a comparison between policies inside one accounting-complete model, under stated assumptions. None is a forecast. The central scenario assumes AI pushes the labour share from 66% to 30% of income by 2048; a milder shift gives smaller effects. The model reproduces the 2019 national accounts, and its first-year fiscal multipliers sit inside the published ranges. Over 2011–2019, though, it tracks history no better than simple benchmarks (Part V). Read the direction of each result and the trade-offs, not the decimals. Every figure is regenerated from the engine by **scripts/build_manifesto.py**.

This is version 1.2. It replaces v1.1 (“Owning the Machine”), whose headline claims did not survive a re-validation of the model: the citizens' dividend overtaking cash, ownership dominating every alternative, and the model being validated against history. Part V lists every change.

Prepared with the AGORA engine · 24 EU economies · November 2026 (v1.2, corrected)

By Branislav Ambroz

Executive summary

If AI raises output while shrinking the share of it paid as wages, three things follow for Europe. AGORA, an open, accounting-complete model of the economies of 24 member states, traces each of them under one stated scenario: the labour share falls from 66% to 30% by 2048, with an investment boom in AI capital. Germany is the worked example; the other 23 members are run the same way.

1. The welfare state's tax base erodes

European welfare states are funded mostly by taxes on wages. With no policy response, German wage-tax revenue falls from 13.8% to 7.1% of GDP by 2048. Taxes on profits rise (5.9% to 8.9%), but not by enough. Across the 24 members the median public debt rises from 55% to 127% of GDP, and 7 pass 150%. This needs no view on basic income: the state taxes the income AI shrinks.

2. Cash and ownership trade consumption now for wealth later

A 40% levy on profits can be paid out as a cash basic income, or taken as shares in a citizens' fund that pays a dividend (Universal Basic Capital, UBC). Cash raises average consumption per person by 13% over no policy; UBC by 1%. UBC cuts the top-10% share of wealth from 56% to 17%, while cash leaves it at 60%, and it ends with the lower income Gini (0.20 against 0.24). The fund's dividend stays below the cash transfer every year to 2048 (€5,324 against €7,739 per person in 2048). Both cost output: GDP ends about 8% below the no-policy path, because both tax the profits that fund investment. Neither form dominates. Choosing between them means choosing between consumption now and ownership later. Across 500 joint draws of the model's uncertain assumptions, UBC is more equal in every draw, and cash gives more consumption in every draw.

3. Europe does not own most of the machine

By 2048 the rent earned by AI capital reaches 36% of German GDP, and 70% of it is paid abroad as licence and cloud fees (US providers hold about 70% of Europe's cloud market). Real GDP rises 10% while real national income falls 13%, and consumption per person ends 23% below 2019. No domestic instrument reaches the part that leaves: a 40% claim on domestic profits reaches at most 25% of the AI rent, and a 3% levy on fees paid abroad reaches 0.3% of the foreign part, because providers pass it on.

What follows

The choice Europe faces is less “cash or capital” than how its welfare state is funded once wages shrink, and who owns the AI it uses. A dividend pooled across the Union narrows the gap between member states by 32% (cash) or 21% (UBC) by 2048, but needs transfers of up to €15,822 per person a year from the richest members.

PART I

The shock: income moves from work to capital

AGORA represents the AI transition as an assumption, not a forecast. From 2019 to 2048 the labour share falls in a straight line from 66% to 30%, and investment in AI capital booms. The profit attributed to AI capital (the AI rent) is split between domestic owners and foreign providers; by default 70% is earned abroad, the US providers' share of the European cloud market (Synergy Research, 2025). The size of this shock is the model's largest driver: across the Monte Carlo, the labour-share fall explains 80% of the variation in German consumption and 73% of the variation in national income.

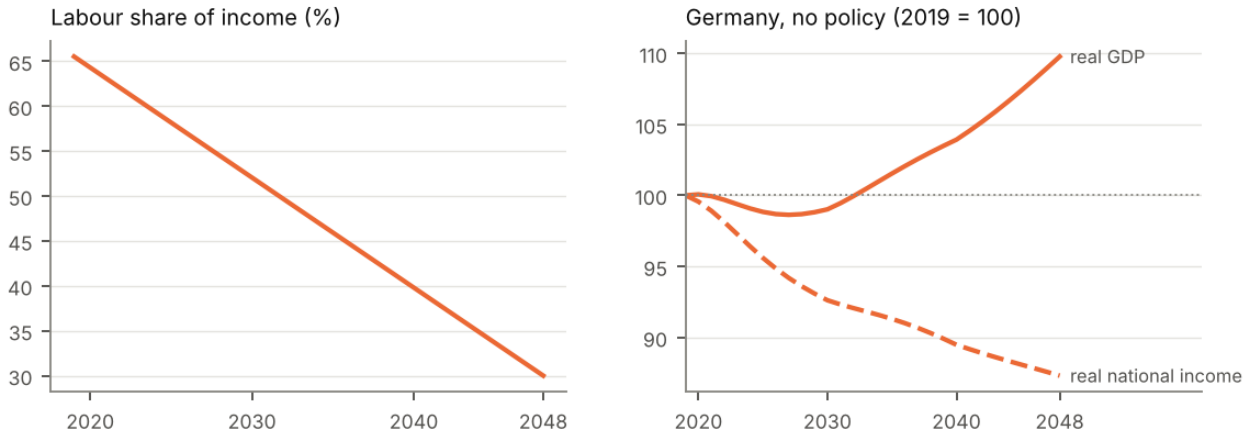


Figure 1. Output keeps growing; national income does not. With no policy, real GDP rises while real national income, which nets out the AI rent paid abroad, falls below its 2019 level.

Source: AGORA engine, AI shift with no policy, Germany, calibrated to 2019 Eurostat and AMECO data. Scenario, not a forecast. Without the AI shift, real GDP grows 46% by 2048 at the 2024 Ageing Report trend.

The shock is not uniform. Using Germany's Eurostat input-output table (FIGARO), AGORA spreads the fall across sectors by their exposure to AI, which is cognitive rather than robotic (OECD 2024 sectoral taxonomy). Labour's share in ICT, finance and business services falls from 45% to 2%; construction moves from 63% to 47%.

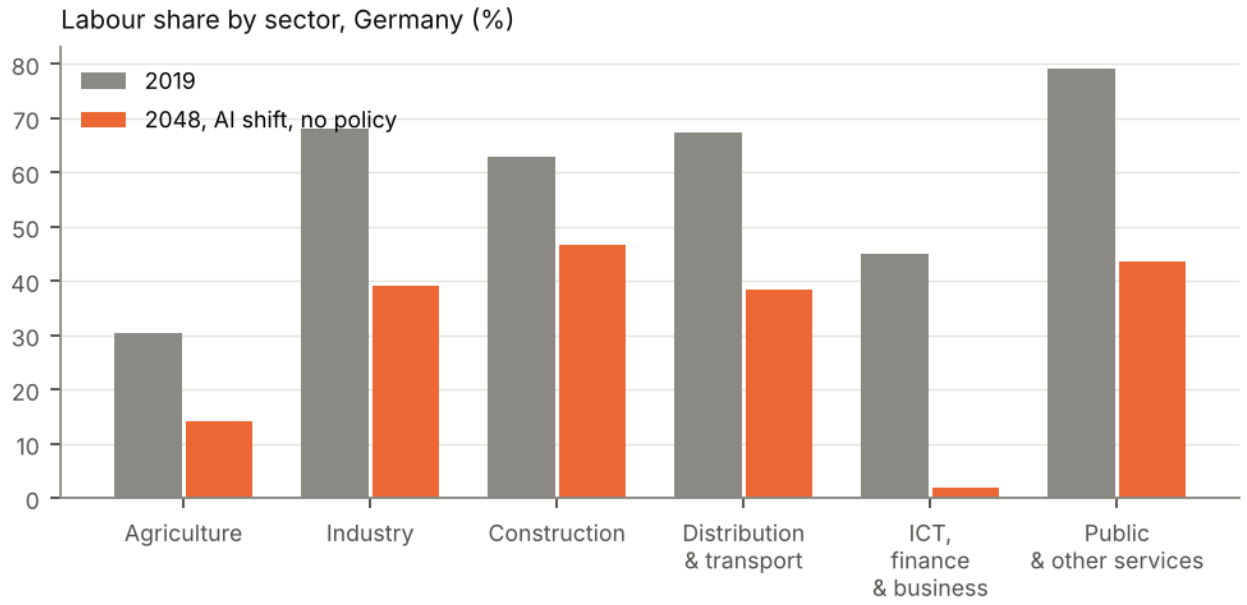


Figure 2. A concentrated shock: office work loses most of its labour share.

Source: AGORA input-output module on real Eurostat FIGARO tables, Germany. Sector value added reconciles to GDP. Labour shares are floored at 2%.

PART II

Finding 1: the welfare state's tax base erodes

Europe's public finances rest on labour. In 2019 the model's taxes on wages (income tax and social contributions) raise 13.8% of German GDP, and taxes on profits 5.9%. As the labour share halves, wage-tax revenue falls to 7.1% of GDP by 2048. Profit taxes rise to 8.9%, because profits grow, but much of the new profit is the AI rent paid abroad, which the German budget never taxes. The government in the model responds as European governments have historically done: it raises tax rates as debt climbs (the wage-tax rate goes from 21.1% to 23.7%). Debt rises anyway.

The pattern holds across the Union. Median wage-tax revenue falls from 13.5% to 7.4% of GDP; public debt rises in 24 of 24 members, from a median of 55% to 127% of GDP, and 7 end above 150% (Figure 3).

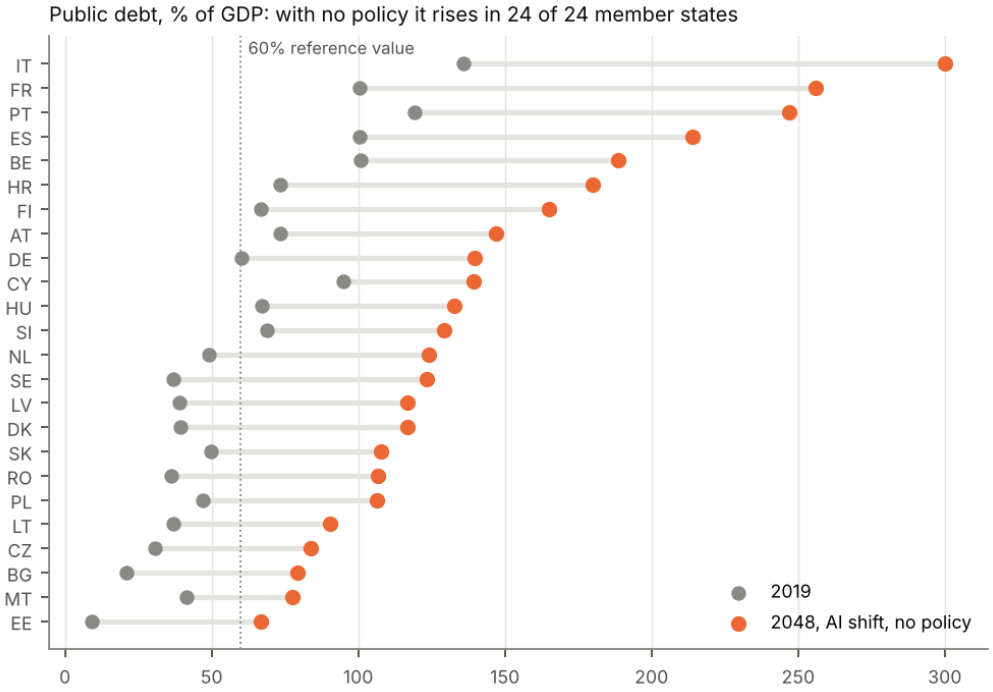


Figure 3. With no policy, public debt rises across the Union.
Source: AGORA engine, AI shift with no policy, 24 members (EL, IE, LU excluded: Part V). The government follows a fiscal rule that raises tax rates as debt rises, and borrows at market rates with a risk premium.

Of the three findings, this one depends least on contested assumptions. It follows from where European taxes fall, and it holds whatever one thinks of basic income or citizens' funds. The levels depend on interest rates and inflation, which the model holds near the ECB's 2% target; the direction does not. Any answer to the AI transition in Europe is also an answer to how the welfare state is funded.

PART III

Finding 2: cash or ownership, consume now or own later

The two forms are compared at the same intensity. Each year 40% of profits is either taxed and paid to every resident as an equal cash income, or converted into new shares owned by a citizens' fund. The fund follows Norway's rule for its sovereign wealth fund: it pays out 3% of its value a year as an equal dividend and keeps the rest invested. Shares are priced at market value, and the fund takes on its share of company debt.

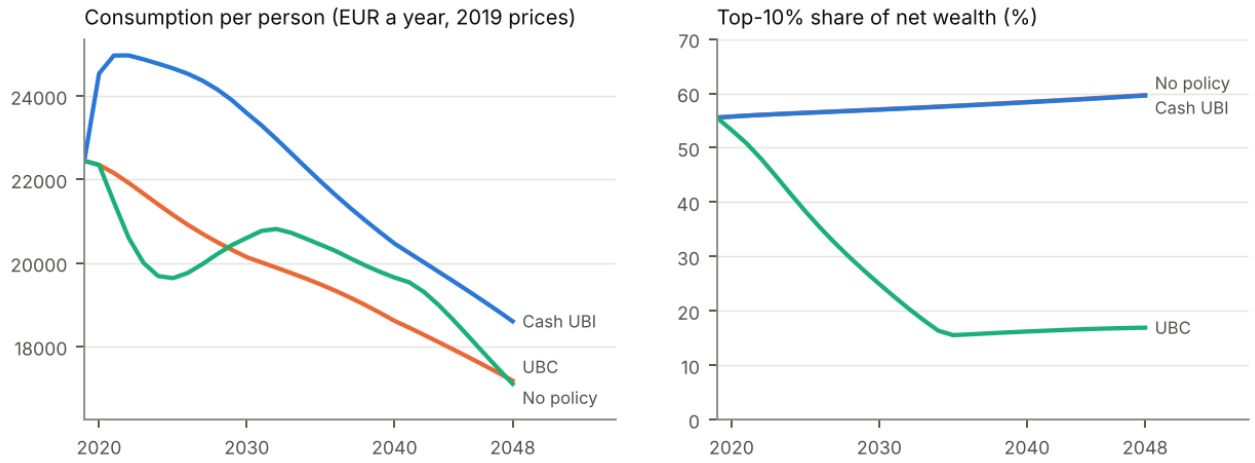


Figure 4. Cash raises consumption; only ownership changes who owns.
Source: AGORA engine, Germany, AI shift, 40% levy paid as cash or 40% claim in shares (fund pays 3% of its value a year). Consumption in 2019 prices.

Figure 4 is the trade-off in one picture. Cash lifts average consumption per person by 13% over 2020–48; UBC by 1%. Wealth runs the other way: with cash, the top-10% share drifts up from 56% to 60%, as it does with no policy; under UBC it falls to 17%, because citizens come to own 84% of domestic capital (61% after ten years), worth €175,197 per person by 2048.

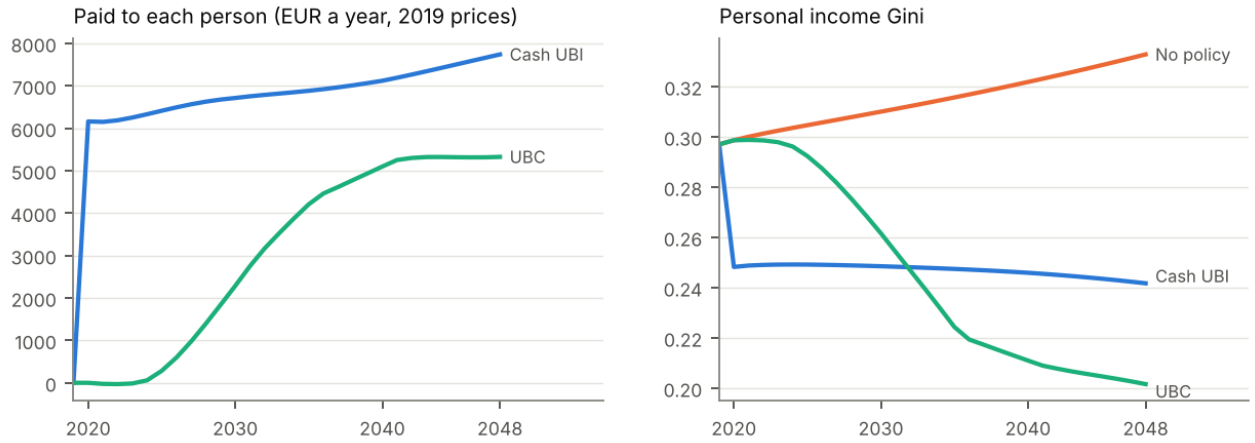


Figure 5. The dividend never catches the cash transfer; the income Gini still ends lower under UBC.
Source: as Figure 4. Payments in 2019 prices.

UBC is in effect a collective savings scheme: most of what the levy raises is kept as capital rather than paid out. Comparing it with cash compares saving with spending. After ten years the cash transfer is €6,675 per person and the dividend €1,836; by 2048 €7,739 against €5,324. Cash cuts poverty faster (after ten years: 10.0% against 13.8%, with 18.2% under no policy). UBC ends lower (4.5% against 7.7% in 2048): under UBC the capital income itself is spread more evenly, while cash adds a flat payment to an unchanged market distribution.

Neither form protects growth

Both forms cost about 8% of 2048 GDP against no policy: real GDP ends 1.10× its 2019 level with no policy, 1.01× with cash and 1.01× with UBC. Investment follows profits, and both take 40% of them. Letting the fund reinvest more of its profit at home does not change this much (Table 1). It moves money from the dividend into capital, the economy ends at most a few per cent larger, and at full reinvestment it is smaller while the Gini rises above cash's.

Table 1. The fund's reinvestment dial (Germany, 2048)

Share of the fund's retained profit reinvested at home	UBC economy (cash economy = 1.00)	UBC Gini, 2048	Dividend per person, 2048
0%	1.00	0.202 (cash 0.242)	€5,324
30%	1.03	0.212 (cash 0.242)	€4,562
60%	1.04	0.226 (cash 0.242)	€3,553
100%	0.97	0.278 (cash 0.242)	€209

Source: AGORA engine, Germany, AI shift, 40% claim in shares.

How robust is this?

The ranking survives the model's uncertainty. Drawing all 19 uncertain assumptions together 500 times for Germany, France and Slovakia (Table 2), UBC is more equal, has less poverty and leaves citizens more wealth in practically every draw, and cash gives more consumption in every draw. On national income the result depends on the country.

Table 2. UBC against cash UBI across 500 joint draws

Share of 500 draws in which UBC beats cash UBI	DE	FR	SK
Lower income Gini, 2048	100%	100%	100%
Lower poverty rate, 2048	100%	98%	100%
More citizen-owned wealth	100%	100%	100%
Higher consumption per person, 2020–48	0%	0%	0%
Higher real national income, 2048	6%	8%	100%

Source: docs/ROBUSTNESS-2026-11.md (scripts/robustness.py), seed 0. The same draw gives both forms the same assumptions.

The size of the inequality gap is far less certain than its direction. 91% of the variation in UBC's Gini comes from one parameter, β, which links the capital share to market inequality. It cannot be estimated from recent data, because European taxes and transfers already erase about 37% of market inequality, hiding the signal. The ranking holds across β's whole plausible range; the gap between the two forms does not.

If a society chooses ownership

- Write the payout rule into law. The split between dividend and reinvestment decides who gains when (Table 1). It should be set by rule, not discretion.
- Keep stakes equal and non-tradable, so ownership cannot re-concentrate through a secondary market, the failure of past voucher privatisations.
- Expect a slow start. At a 3% payout the dividend stays small for a generation. Pairing a fund with a cash floor is a design option the model does not yet test.

PART IV

Finding 3: Europe does not own most of the machine

The AI rent is large and mostly leaves. By 2048 it reaches 36% of German GDP, and 25% of GDP is paid to foreign providers. That is why, in Figure 1, real GDP rises 10% while real national income falls 13%. The fees leave the national income that households and the state live on: consumption per person ends 23% below 2019 with no policy, and below 2019 under cash and UBC too (Figure 4). Taxes on domestic profits and claims on domestic shares reach only the domestic part. The 40% claim in shares reaches 25% of the AI rent; the 40% cash levy 12%.

Table 3 runs six real 2026 proposals against the two 40% benchmarks. None reaches much of the foreign rent. A 3% levy on fees paid abroad reaches 0.3% of it, because providers raise prices by about the levy; voluntary pledges by AI companies, in their best case, reach 13%. The real proposals are also small: none moves national income by more than a few per cent.

Table 3. Real proposals against no policy (Germany, AI shift)

Policy	Real national income, 2048 (2019 = 1)	Consumption per person, 2020–48	Gini, 2048	Public debt, 2048 (% GDP)	Share of foreign AI rent reached
No policy (AI shift)	0.87	€19,639	0.333	140	0.0%
One-time stock levy on AI firms (Sanders)	0.87	€19,857	0.314	125	0.0%
Voluntary AI pledges, best case (Anthropic-style)	0.90	€20,283	0.317	135	13.0%
Borrowing-financed citizens' fund	0.90	€20,014	0.327	168	0.0%
Child accounts (Fruehstart-Rente)	0.87	€19,638	0.333	140	0.0%
Levy on AI fees paid abroad (3%)	0.87	€19,565	0.333	131	0.3%
Cash UBI on a broad tax base	0.86	€20,382	0.281	139	0.0%
Benchmark: 40% capital levy, cash UBI	0.80	€22,177	0.242	137	0.0%
Benchmark: 40% claim in shares, fund pays 3%	0.78	€19,827	0.202	131	0.0%

Source: AGORA proposals library (proposals.py), every run consistency-gated. Each proposal runs at its announced size; the two benchmarks are far larger than any of them.

The model can say that the foreign part is out of reach of domestic taxes and claims. Whether Europe can build its own AI providers or take stakes in foreign ones is a question of industrial policy, outside the model. But it is where most of the money is.

Between member states: pooling

A purely national dividend lets richer members pay bigger dividends, so the gap between countries persists. Pooling the levy across the Union and paying an equal European dividend narrows the between-country Gini by 32% under cash and 21% under UBC by 2048 (Figure 6). After ten years cash pooling has narrowed it 25%, while UBC pooling has widened it 5%. The transfers are large: under cash pooling MT pays €15,822, DK pays €11,108, NL pays €10,150 per person a year by 2048, and BG receives €7,483, RO receives €6,083, LV receives €5,707. The arithmetic works; the politics would be hard.

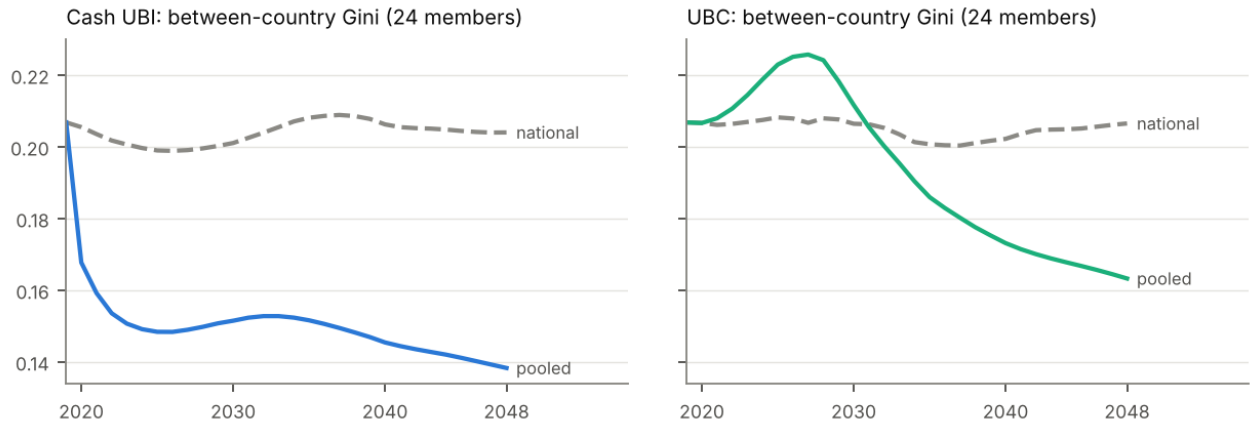


Figure 6. Pooling narrows the gap between member states under both forms.
Source: AGORA multi-region module, 24 members, 40% levy, transfers routed through each country's accounts and iterated to a fixed point.

PART V

What the model can and cannot tell you

History test

The model was started in 2010 from 2010 data only, fed the government spending, exports, wage share and interest rates that actually happened, and left to produce everything else to 2019. It was scored against two naive benchmarks: a random walk, and a static multiplier given the same inputs. The bar, set before the run, was to beat both for most countries and variables. It does not: the engine beats both in 62 of 168 cases (37%). Inflation is its best variable and the current account its worst.

Table 4. Median error, 2011–2019, 24 members

Variable	Engine error	Random walk	Static multiplier	Engine beats both
Real GDP	3.9%	4.0%	5.5%	10 of 24
Real consumption	5.0%	5.5%	5.7%	8 of 24
Real private investment	12.2%	17.3%	11.9%	5 of 24
Inflation	1.1 pp/yr	1.8 pp/yr	1.8 pp/yr	17 of 24
Public debt / GDP	8.4 pp	7.3 pp	7.6 pp	8 of 24
Current account / GDP	2.8 pp	2.6 pp	2.0 pp	2 of 24
Private-sector balance / GDP	2.8 pp	2.6 pp	2.5 pp	12 of 24

Source: docs/VALIDATION-2026-11.md (scripts/validate.py). Errors are changes since 2010.

Its short-run responses are closer to the evidence. A stimulus of 1% of GDP raises GDP by 0.82 per euro of public spending in the first year, 0.42 for a cut in taxes on wages and 0.22 for a cut in taxes on profits, all inside the ranges of European central-bank models. In the second year the responses are above them (spending 1.16), because the model has no central-bank reaction; read its short-run effects as upper bounds.

Limits

- The AI shock is assumed, and its size drives most results (Part I).
- No exchange rates and no monetary reaction. Surplus countries keep accumulating foreign assets; the current account is the weakest variable in the history test.
- Inflation is anchored near 2%. Over 2011–19 actual inflation was lower, so the model under-predicted debt in most countries.

- The 70% foreign share of the AI rent is proxied by the US share of Europe's cloud market (Monte-Carlo range 50–90%).
- EL, IE, LU are excluded: Greek households spent more than their income in 2019 under the model's calibration, and multinational accounting distorts Irish and Luxembourg GDP.
- Outside the model: the politics of a capital levy, fund governance and capture, capital flight, and whether the AI shock comes at all.

What changed from v1.1

Between October and November 2026 the model gained capacity limits and prices, credit constraints for owners, foreign ownership of capital and of the AI rent, market prices for shares, a banking sector, existing loans, each country's own capital stock and trend growth, and gradual consumption. Each step made the effects smaller and turned clear wins into trade-offs. v1.1 claims that do not survive:

- “The citizens' dividend overtakes cash around year 11.” Retracted: it stays below the cash transfer every year to 2048.
- “UBC dominates both cash and inaction; doing nothing is off the frontier.” Retracted: no policy and cash UBI at every rate are on the trade-off frontier.
- “With reinvestment the UBC economy ends larger than the cash economy.” Retracted: at most 1.04× (Table 1).
- “Back-tested to ~2.3% mean GDP error; the model is validated.” Retracted: see Table 4.
- “Pooling roughly halves the European gap.” Changed: 32% (cash) and 21% (UBC).
- “UBC eliminates poverty.” Changed: it falls to 4.5%, against 7.7% under cash.

What survives: only ownership moves the concentration of wealth, and UBC is more equal than cash in every draw. Every published finding and its status is listed in **docs/FINDINGS-LEDGER.md**.

The bottom line

If AI moves income from work to capital on anything like this scale, Europe's welfare states lose their tax base, and much of the new income leaves the continent. Cash and citizen ownership answer different questions: cash protects consumption now, ownership changes who holds wealth later, and both cost some growth. The model cannot choose between them; that is a choice of values. It can show what each choice costs, in an open model anyone can rerun and change.

APPENDIX

Method, data and reproducibility

The engine. AGORA is a stock-flow-consistent model: every euro is someone's asset and someone else's liability, and every scenario must pass a consistency gate (sector balances sum to zero, stocks match accumulated flows) before any result is reported. It combines a macro core with capacity limits, prices, a banking sector and a government with a fiscal rule; a household income and wealth distribution; a sectoral input-output layer; foreign ownership of capital and of the AI rent; a citizens'-fund mechanism; and a multi-country layer for the Union. Each run is also checked against plausibility laws (no insolvent households, investment below 40% of GDP, Gini above 0.15).

Calibration. Each of the 24 members is calibrated to its 2019 national accounts, its own capital stock, debt, interest rates, foreign position and existing loans, and grows at its 2024 Ageing Report trend without the AI shift. Every parameter is sourced or labelled as an assumption with a Monte-Carlo range.

Data sources (through DBnomics unless noted)

Layer / quantity	Source
GDP, consumption, investment, trade; history 1995–2019	Eurostat national accounts (nama_10_gdp, gov_10a_main, nasa_10_ki)
Labour share; capital stock and depreciation	AMECO (ALCD2; OKND, PIGT, UKCT, UVGD)
Public debt, deficit, interest	Eurostat gov_10dd_edpt1
Interest rates, loan and deposit rates	AMECO (ISN, ILN); ECB MFI interest rates
Existing loans of firms and households	Eurostat financial accounts (nasa_10_f_bs, F4)
Foreign ownership of capital; foreign positions	Eurostat nasa_10_nf_tr (D42+D43 with the rest of the world); tipsii10
Capacity utilisation	European Commission business surveys (ei_bsin_q_r2)
Trend growth to 2070	2024 Ageing Report, European Commission Institutional Paper 257
Propensities to consume	Onaran & Obst (2016); Jappelli & Pistaferri (2014); Fagereng, Holm & Natvik (2021)
Income and wealth distribution	Eurostat ilc_di12; OECD IDD and Wealth Distribution Database
Sectoral structure; AI exposure	Eurostat FIGARO (naio_10_cp1700); OECD (2024) sectoral taxonomy of AI intensity
Foreign share of the AI rent	Synergy Research (2025), European cloud market shares

Reproducibility. The full test suite (423 tests) passes, and every number in this document is recomputed from the gated engine by **scripts/build_manifesto.py**, which refuses to build if any scenario fails the gate. The history test and the Monte Carlo are read from files written by **scripts/validate.py** and **scripts/robustness.py**.

AGORA: a sandbox for the economics of abundance. Scenario comparisons under explicit assumptions; not investment, legal or policy advice.